

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461233.884 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.159 +/- 0.047
Transit depth (Rp/Rs)^2	2.54 +/- 1.5 [%]
Orbital Inclination (inc)	89.65 +/- 2.35
Airmass coefficient 1 (a1)	0.94 +/- 0.5
Airmass coefficient 2 (a2)	0.03 +/- 0.12
Scatter in the residuals of the lightcurve fit is	53.53 %
Best Comparison Star	#3 - [392, 242]
Optimal Aperture	2.24
Optimal Annulus	3.09
Transit Duration (day)	0.104 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.159 ± 0.047 vs 0.1202 (deviation 0.83σ)
Duration fit vs published	0.104 d vs 0.1225 d (deviation 1.42σ)
Mid-time O–C	-3.9 min
Depth SNR	3.4
Residual scatter	53.53 %

Light curve

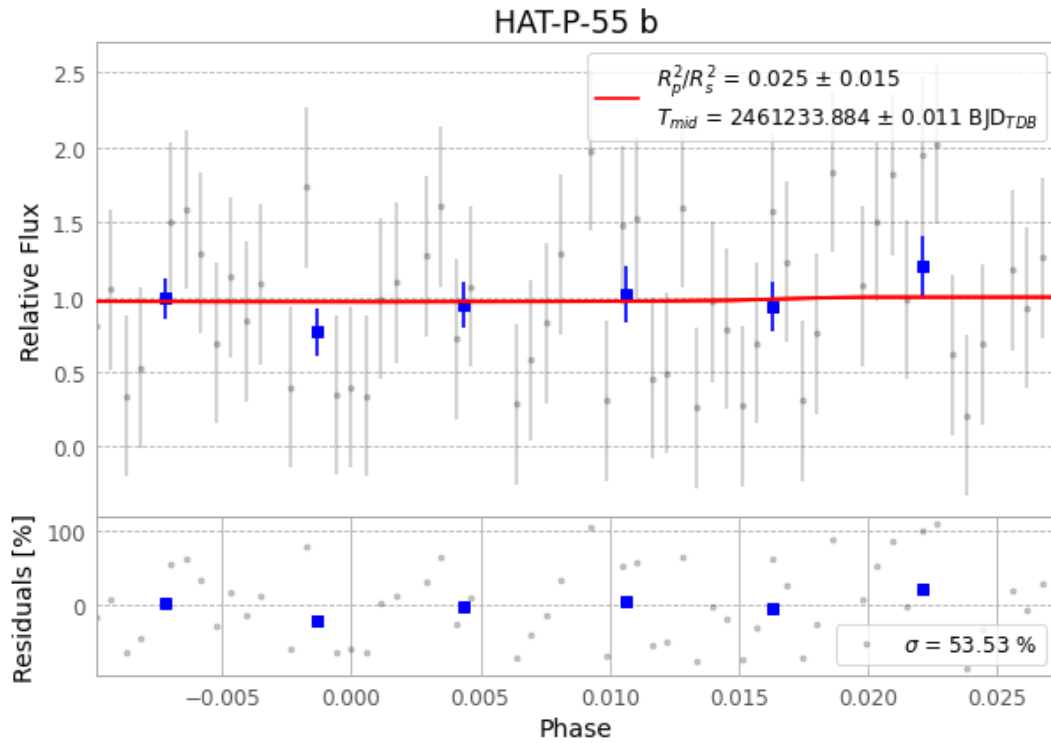


Figure 1 — FinalLightCurve_HAT-P-55 b_2026-07-12.png (SHA-256 4edf42c4346b49c8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [504, 257], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b55e28cbe063ab83...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

65 FITS frames (43 MB), HATP-55260712081515.FITS → HATP-55260712112715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-55-260712.log (ee9c99c5c698...), FinalLightCurve_HAT-P-55 b_2026-07-12.png (4edf42c4346b...), FinalLightCurve_HAT-P-55 b_2026-07-12.csv (a29dc965547b...)

Compute resources

Wall time 143.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 00:51Z.